Sensors Are Messengers

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Recall and choose your starting point

> Record your starting explanation for Sensors Are Messengers before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Input, Process, Output

Without opening the old lesson, explain one key idea from **Input**, **Process**, **Output**. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Practise with fading help

> Complete the supported Sensors Are Messengers trace and record the first step you can explain without help.

Use W2: Trace a boundary for **Sensor**s and **measurement**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

Individual reasoning check

> Check the proposed Sensors Are Messengers explanation against the supplied facts before discussing it.

Use the worked case for **Sensor**s and **measurement**. Proposed explanation: “The more stable device must be accurate”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Investigate a new measurement dispute

A classroom sound display near a speaker reads higher than one near the door. A student declares the door device broken. Propose a safe comparison using normal classroom sound.

Predict one result consistent with location explaining the difference and one result that would leave a device difference worth investigating.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Sensors Are Messengers target to a trace and a justified conclusion.

Two light **sensor**s report 40 and 80 in different rooms. Their settings and **unit**s are unknown. Can you conclude the second room is twice as bright? Design a fair comparison without using a camera or personal **data**.

Solve the given case

> Solve the supplied Sensors Are Messengers case and show the deciding evidence.

A light **sensor** reads 15 in a cupboard and 90 near a window. A microphone records a clap.

> Which measures a light-related **signal**? Can either directly measure happiness?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A station reports light level 18 and temperature 22 degrees C. Which **sensor** produced each reading? Can either reading alone tell whether a child feels comfortable? Explain.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 3 Test and reflect

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	

Test record	My evidence
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Sensors Are Messengers adds to Helpful Classroom Sorter and keep the evidence.

Mark which **sensor**, if any, would collect each Classroom Sorter **input**; prefer direct human entry when a **sensor** is unnecessary.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.

- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Two light **sensor**s report 40 and 80 in different rooms. Their settings and **unit**s are unknown. Can you conclude the second room is twice as bright? Design a fair comparison without using a camera or personal **data**.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Repair an incomplete record

> Repair an incomplete record. Apply the supplied Sensors Are Messengers facts and keep your reasoning.

A notebook entry says 24 at lunchtime. Rewrite it as a useful air-temperature record by stating which missing fields must be collected. Do not invent values that were not observed.

My reasoning and evidence

W2 Trace a boundary

> Trace a boundary. Apply the supplied Sensors Are Messengers facts and keep your reasoning.

A warning rule is recorded temperature greater than 28 degrees C. State its **output** for 27, 28 and 29 degrees C. Explain why an off warning does not prove that the **sensor** is working correctly.

My reasoning and evidence

W3 Compare without choosing a winner

> Compare without choosing a winner. Apply the supplied Sensors Are Messengers facts and keep your reasoning.

At the same three times, A records 18, 19 and 18 degrees C and B records 21, 22 and 21 degrees C in the same location. Calculate the paired differences. What is established, and what further **evidence** would help judge accuracy?

My reasoning and evidence

W4 Investigate a new measurement dispute

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Investigate a new measurement dispute. Apply the supplied Sensors Are Messengers facts and keep your reasoning.

A classroom sound display near a speaker reads higher than one near the door. A student declares the door device broken. Propose a safe comparison using normal classroom sound.

Predict one result consistent with location explaining the difference and one result that would leave a device difference worth investigating.

My reasoning and evidence

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W5 Limit a system's conclusion

> Limit a system's conclusion. Apply the supplied Sensors Are Messengers facts and keep your reasoning.

An air-temperature **sensor** reports 29 degrees C and a rule turns on a warning. Someone writes The plant is thirsty.

Explain the unsupported step and name information more directly relevant to deciding whether watering is needed.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Sensors Are Messengers

> Two fictional light sensors disagree in a classroom. Compare them fairly rather than declaring one broken. Explain the labelled step before checking the changed condition.

Two fictional light **sensor**s disagree in a classroom. Compare them fairly rather than declaring one broken.

On the same shelf, simultaneous readings are A: 40,50,60 and B: 45,55,65 in declared toy light **unit**s.

Neither device has been compared with a trusted reference.

AI ACADEMY / AI-1/3

Simultaneous light readings

	Sensor A	Sensor B	B minus A
Time 1	40	45	5
Time 2	50	55	5
Time 3	60	65	5

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/3

Sensors Are Messengers

Start with the supplied facts
Follow each arrow. Ask which fact allows the next step.

1. Same shelf

Place both devices together and allow them to settle.

2. Paired readings

Record three matching times: (40,45), (50,55), (60,65).

3. Compare gaps

B minus A is 5, 5 and 5 units.

4. Check a reference

A trusted reference is needed to decide which reading is closer.

CHANGE ONE THING
Move only B into brighter light.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why can we calculate the earlier gap but not predict an exact new B reading?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Save sorter **measurement**s with device, location and time so unusual **input**s can be checked before decisions.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Save sorter **measurement**s with device, location and time so unusual **input**s can be checked before decisions.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Two fictional light **sensor**s disagree in a classroom. Compare them fairly rather than declaring one broken.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? A trusted reference is needed to decide which reading is closer.

1. Compare gaps

2. Check a reference

3. Paired readings

4. Same shelf

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check a reference A trusted reference is needed to decide which reading is closer. A repeated difference of 5 supports investigating an offset between devices.

Matching time and location removes two obvious comparison problems. It does not show which device is correct: both could have errors. Preserve paired readings instead of adjusting them silently.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Move only B into brighter light. Which conclusion is supported by the supplied facts?

1. The next gap could change for reasons other than a difference in device accuracy.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The next gap could change for reasons other than a difference in device accuracy.

> The comparison now changes location as well as device; there is no supplied reading from the new location.

Repair a missing building block

Review **Input**, **Process**, **Output** (ai-1/2). Goal: Trace information through a system without confusing information with a physical action. Short practice: Use the richer record to find blue pencils in Box B.

Explain why total alone was insufficient.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

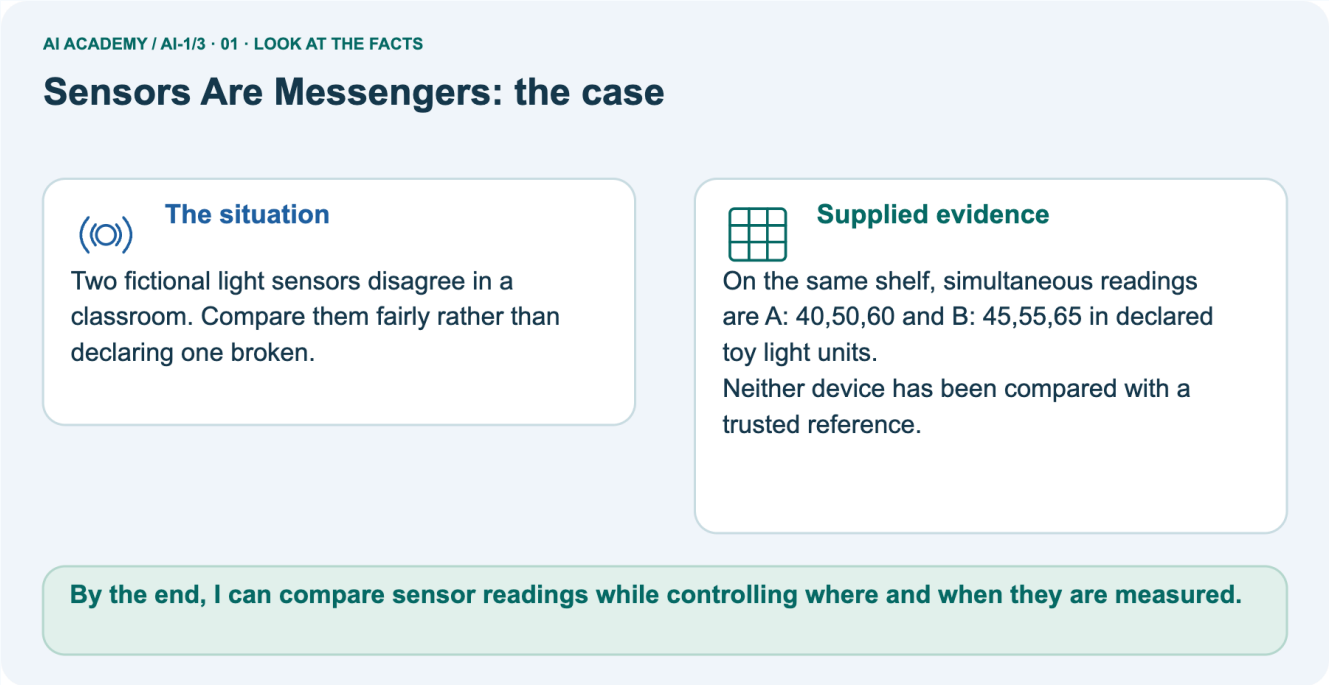
> **What I repaired, the evidence I kept and the next test**
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> | |
> |---|
> | |
> | |
> | |
> | |

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Sensors Are Messengers picture from its labels, then record a relevant limit.

01. Sensors Are Messengers: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

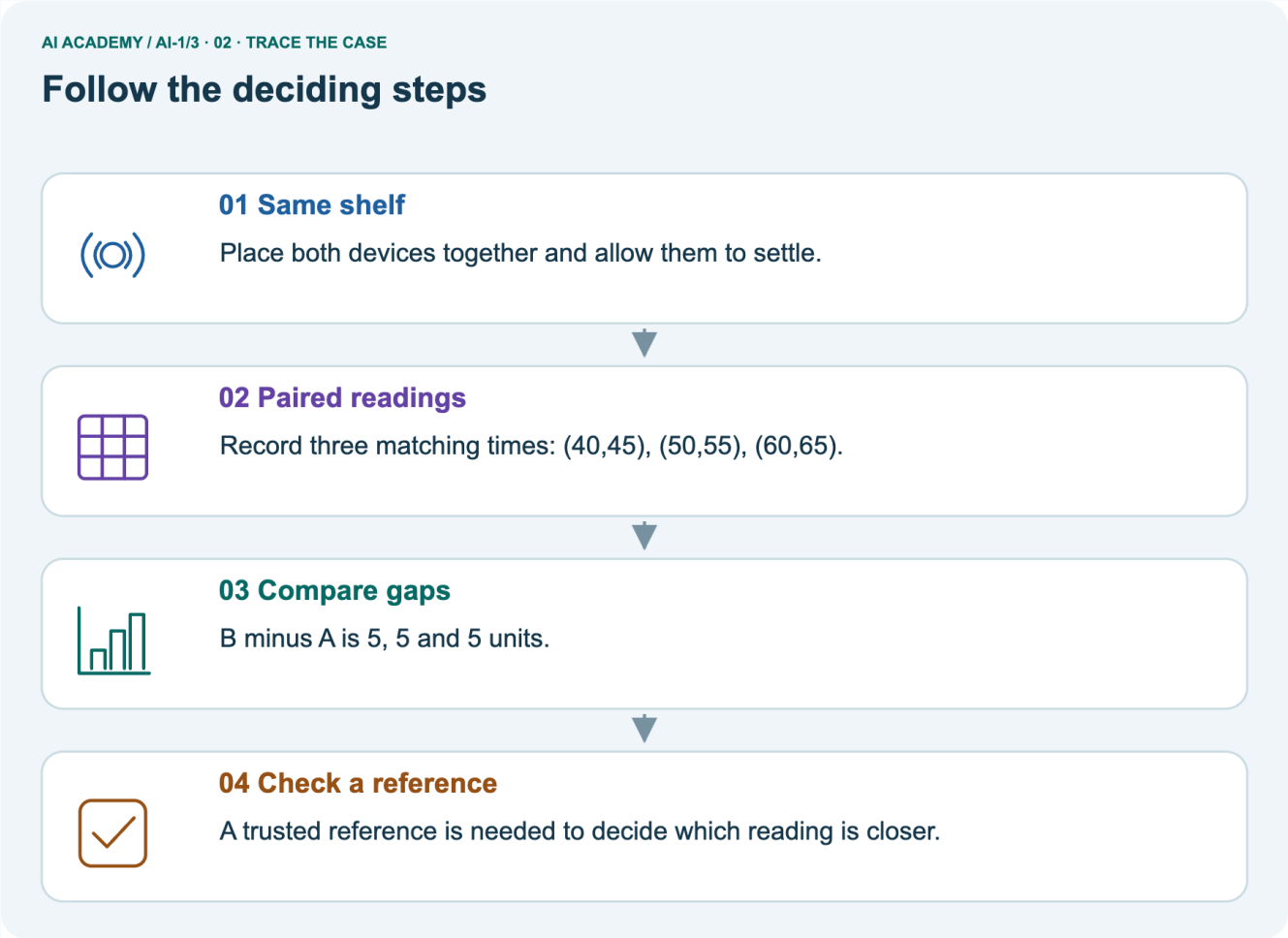
Two fictional light **sensor**s disagree in a classroom. Compare them fairly rather than declaring one broken. Supplied facts: On the same shelf, simultaneous readings are A: 40,50,60 and B: 45,55,65 in declared toy light **unit**s.

Neither device has been compared with a trusted reference.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



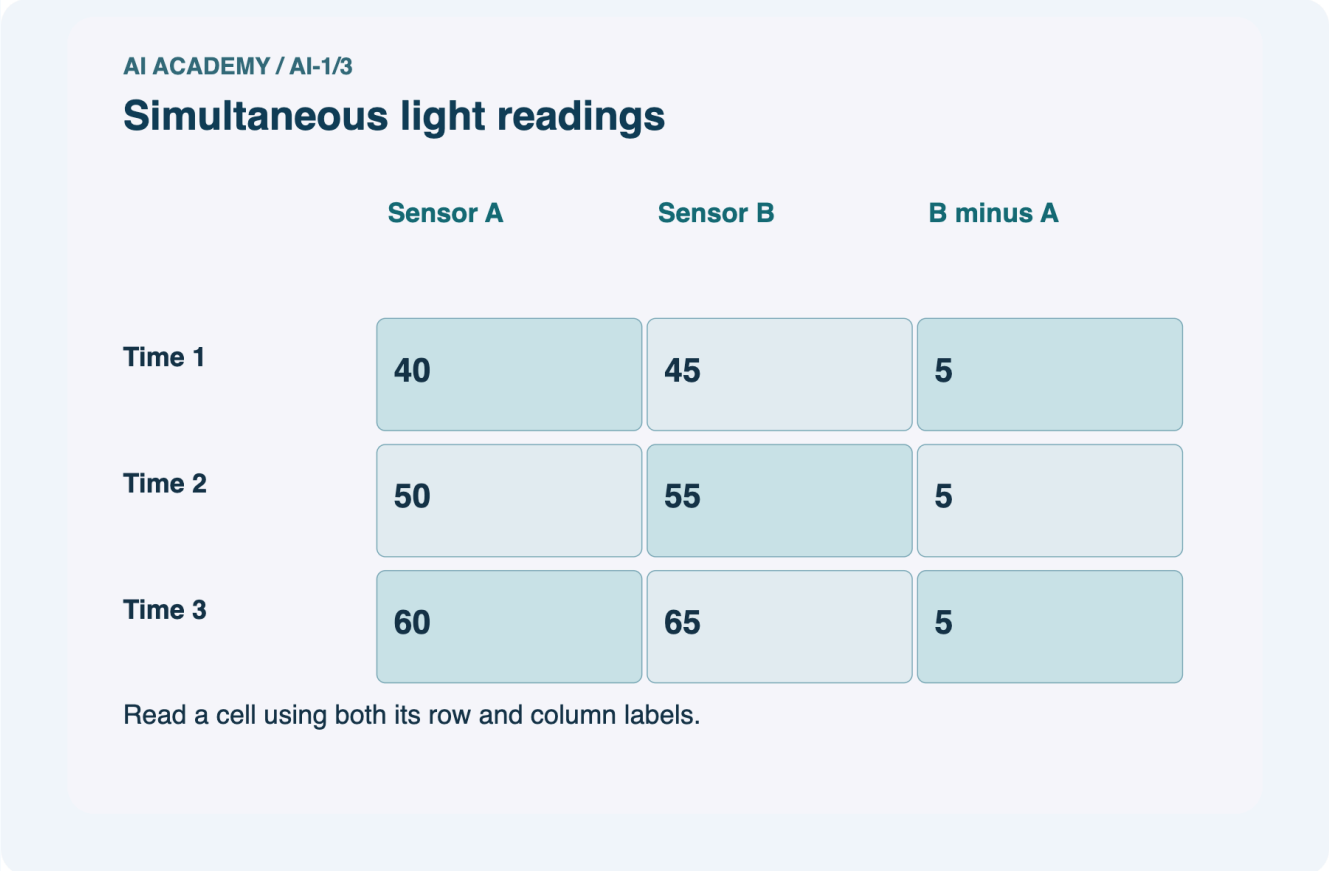
Follow the labelled picture with the matching explanation. Open the vector diagram

Same shelf: Place both devices together and allow them to settle. → Paired readings: Record three matching times: (40,45), (50,55), (60,65). → Compare gaps: B minus A is 5, 5 and 5 **unit**s. → Check a reference: A trusted reference is needed to decide which reading is closer.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Simultaneous light readings



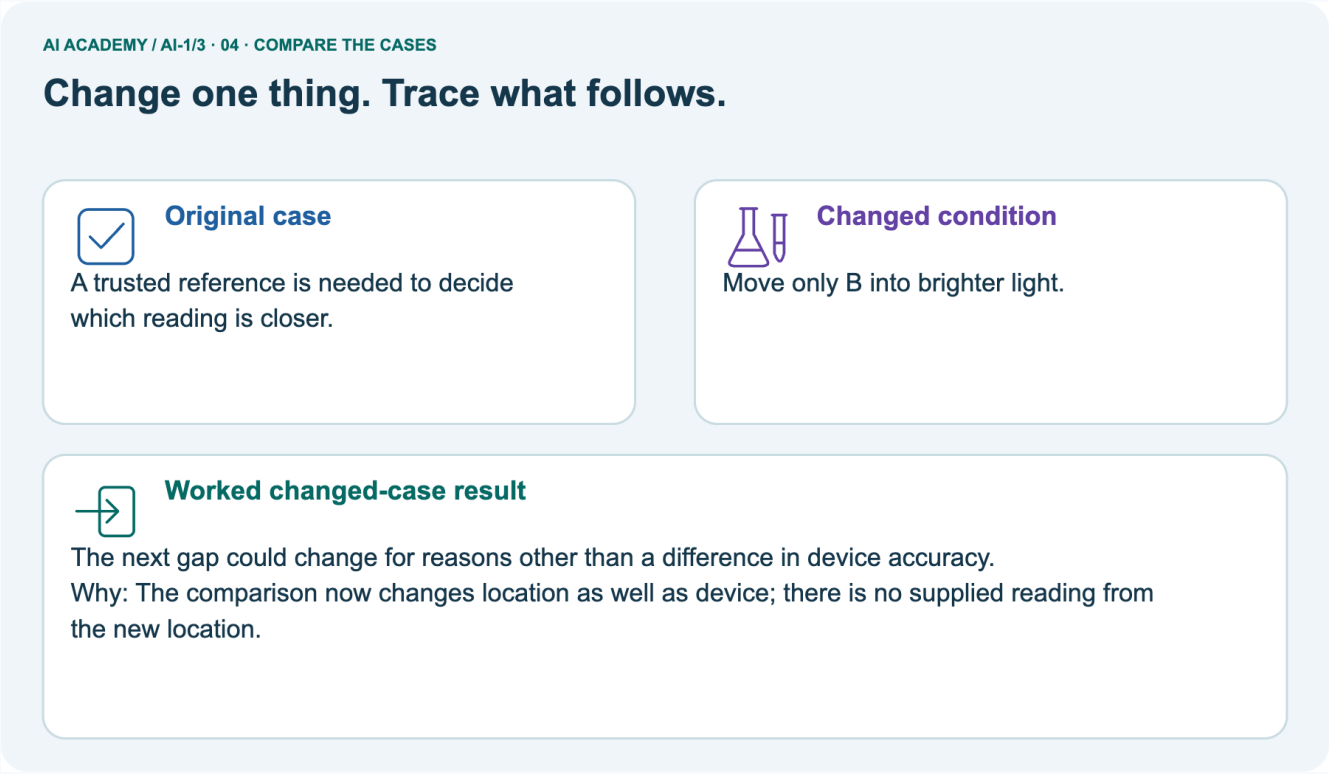
Follow the labelled picture with the matching explanation. Open the vector diagram

Simultaneous light readings. Time 1: **Sensor** A = 40, **Sensor** B = 45, B minus A = 5. Time 2: **Sensor** A = 50, **Sensor** B = 55, B minus A = 5. Time 3: **Sensor** A = 60, **Sensor** B = 65, B minus A = 5.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: A trusted reference is needed to decide which reading is closer. Changed condition: Move only B into brighter light.

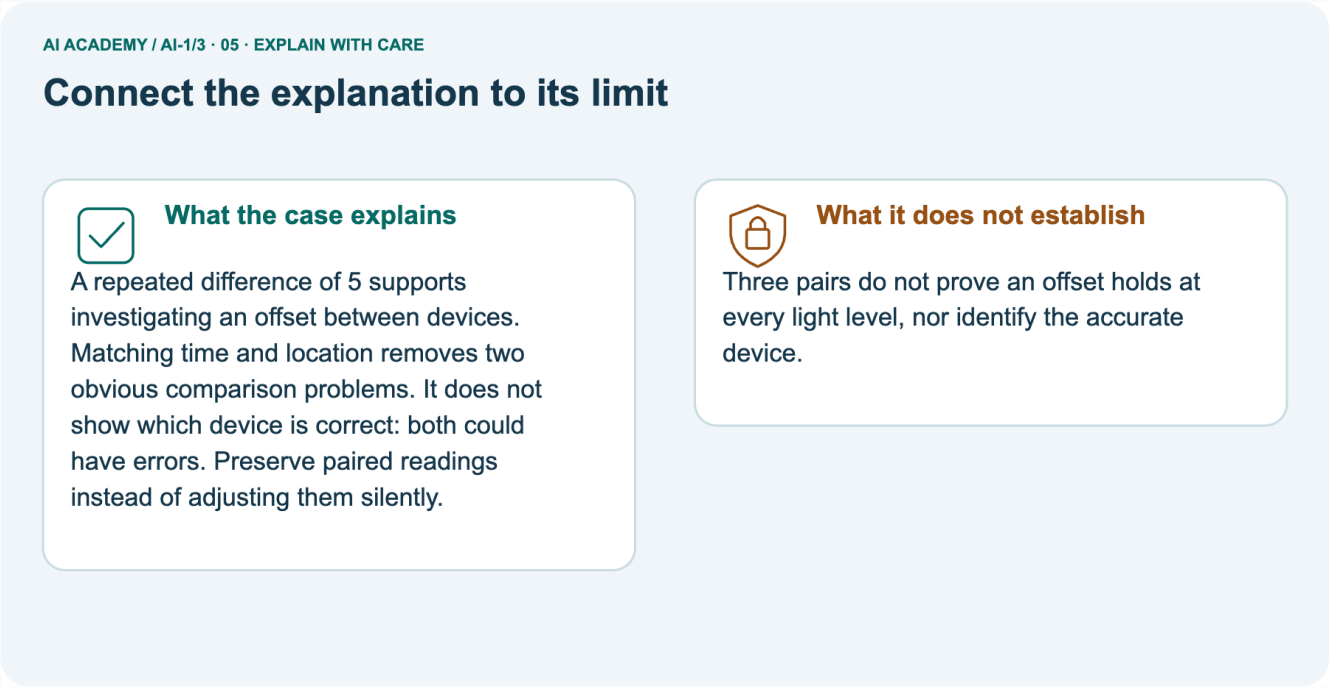
Expected result: The next gap could change for reasons other than a difference in device accuracy. Reason: The comparison now changes location as well as device; there is no supplied reading from the new location.

Predict before checking the worked changed-case result. Name what changed and what the

supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

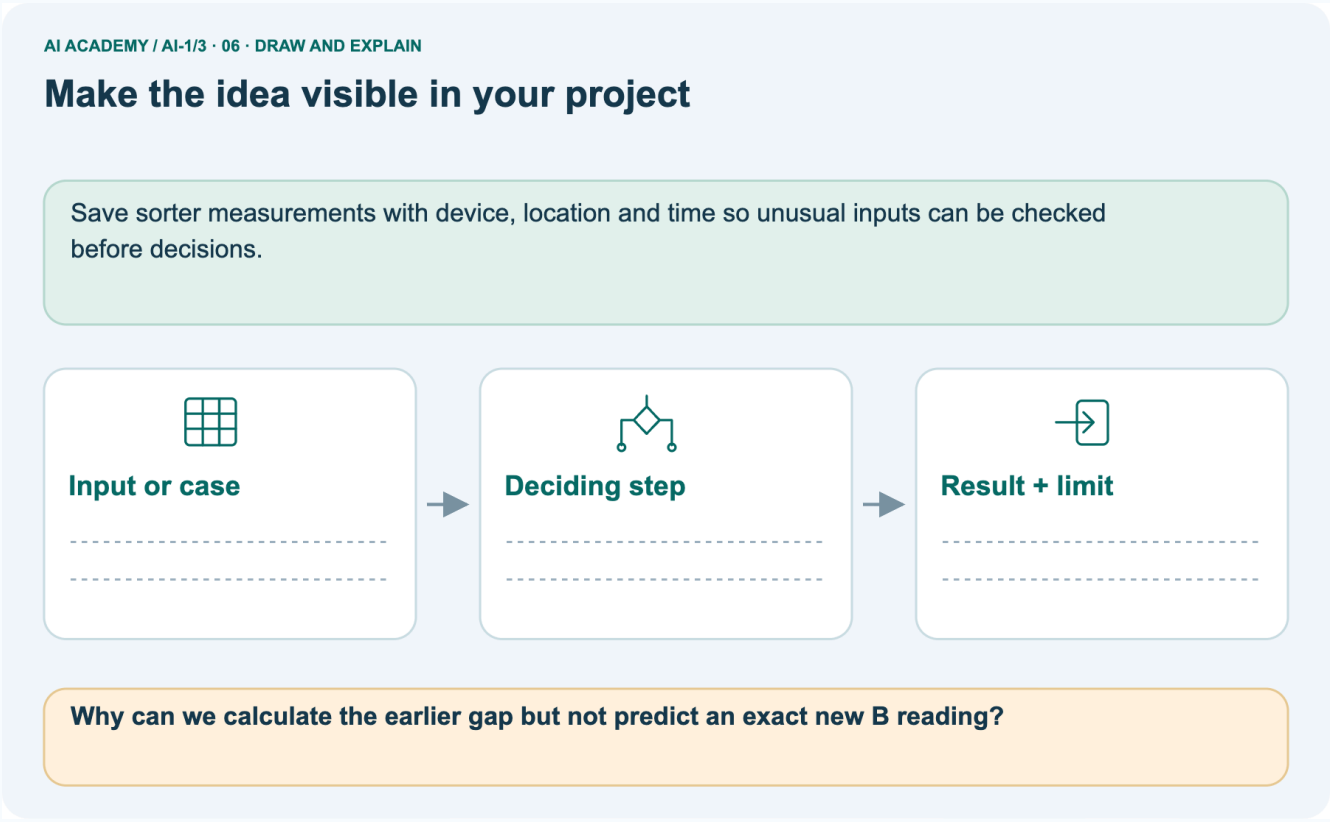
A repeated difference of 5 supports investigating an offset between devices. Matching time and location removes two obvious comparison problems. It does not show which device is correct: both could have errors.

Preserve paired readings instead of adjusting them silently. Limit: Three pairs do not prove an offset holds at every light level, nor identify the accurate device.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Save sorter **measurement**s with device, location and time so unusual **input**s can be checked before decisions. Draw your **input**, deciding step, result and limit.

Supported question: Why can we calculate the earlier gap but not predict an exact new B reading?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.